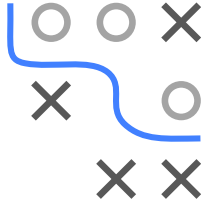
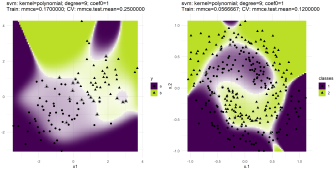
Supervised Learning



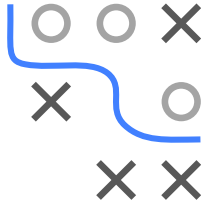
Nonlinear Support Vector Machines
The Polynomial Kernel



Learning goals

- Know the homogeneous and non-homogeneous polynomial kernel
- Understand the influence of the choice of the degree on the decision boundary

HOMOGENEOUS POLYNOMIAL KERNEL I



$$k(\boldsymbol{x}, \tilde{\mathbf{x}}) = (\boldsymbol{x}^T \tilde{\mathbf{x}})^d, \text{ for } d \in \mathbb{N}$$

The feature map contains all monomials of exactly order d.

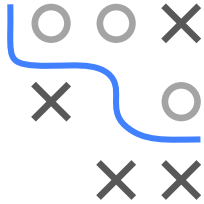
$$\phi(\boldsymbol{x}) = \left(\sqrt{\binom{d}{k_1, \ldots, k_p}} x_1^{k_1} \ldots x_p^{k_p} \right)_{k_i \geq 0, \sum_i k_i = d}$$

That $\langle \phi(\boldsymbol{x}), \phi(\tilde{\boldsymbol{x}}) \rangle = k(\boldsymbol{x}, \tilde{\mathbf{x}})$ holds can easily be checked by simple calculation and using the multinomial formula

$$(x_1 + \ldots + x_p)^d = \sum_{k_i \geq 0, \sum_i k_i = d} \binom{d}{k_1, \ldots, k_p} x_1^{k_1} \ldots x_p^{k_p}$$

The map $\phi(\boldsymbol{x})$ has $\binom{p+d-1}{d}$ dimensions. We see that $\phi(\boldsymbol{x})$ contains no terms of "lesser" order, so, e.g., linear effects. As an example for $p = d = 2$: $\phi(\boldsymbol{x}) = (x_1^2, x_2^2, \sqrt{2} x_1 x_2)$.

NONHOMOGENEOUS POLYNOMIAL KERNEL I



$$k(\boldsymbol{x}, \tilde{\mathbf{x}}) = (\boldsymbol{x}^T \tilde{\mathbf{x}} + b)^d, \text{ for } b \geq 0, d \in \mathbb{N}$$

The maths is very similar as before, we kind of add a further constant term in the original space, with

$$(\boldsymbol{x}^T \tilde{\mathbf{x}} + b)^d = (x_1 \tilde{x}_1 + \ldots + x_p \tilde{x_p} + b)^d$$

The feature map contains all monomials up to order d.

$$\phi(\boldsymbol{x}) = \left(\sqrt{\binom{d}{k_1, \ldots, k_{p+1}}} x_1^{k_1} \ldots x_p^{k_p} b^{k_{p+1}/2} \right)_{k_i \geq 0, \sum_i k_i = d}$$

The map $\phi(\boldsymbol{x})$ has $\binom{p+d}{d}$ dimensions. For $p = d = 2$:

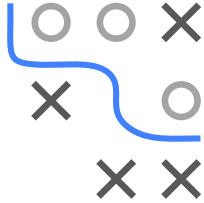
$$(x_1 \tilde{x}_1 + x_2 \tilde{x}_2 + b)^2 = x_1^2 \tilde{x}_1^2 + x_2^2 \tilde{x}_2^2 + 2x_1 x_2 \tilde{x}_1 \tilde{x}_2 + 2bx_1 \tilde{x}_1 + 2bx_2 \tilde{x}_2 + b^2$$

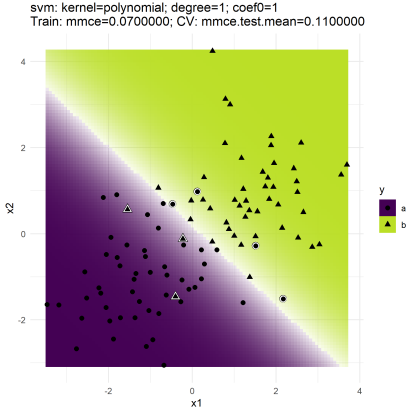
Therefore,

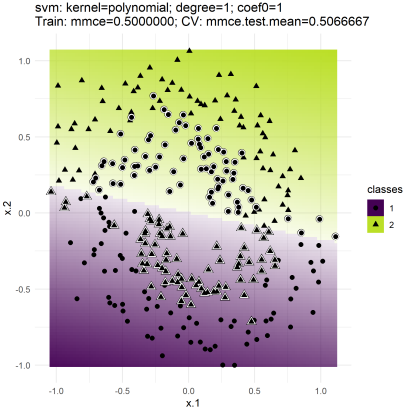
$$\phi(\boldsymbol{x}) = (x_1^2, x_2^2, \sqrt{2} x_1 x_2, \sqrt{2b} x_1, \sqrt{2b} x_2, b)$$

POLYNOMIAL KERNEL I

Degree $d = 1$ yields a linear decision boundary.

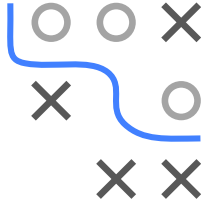


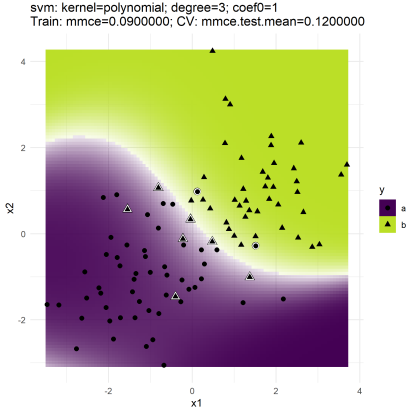


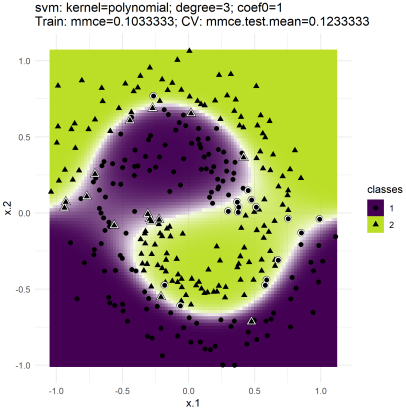


POLYNOMIAL KERNEL II

The higher the degree, the more nonlinearity in the decision boundary.

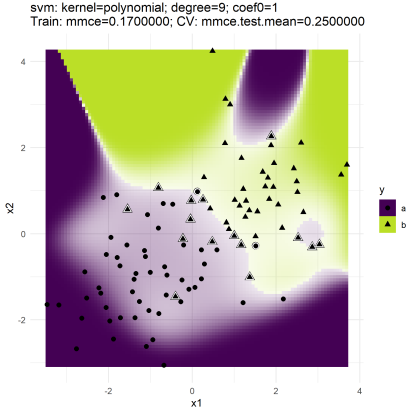


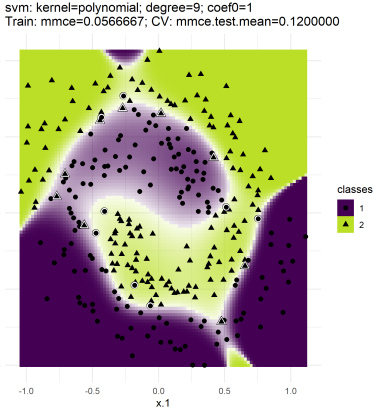


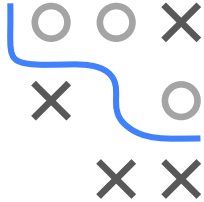


POLYNOMIAL KERNEL III

The higher the degree, the more nonlinearity in the decision boundary.







POLYNOMIAL KERNEL IV

For $k(\boldsymbol{x}, \tilde{\boldsymbol{x}}) = (\boldsymbol{x}^\top \tilde{\boldsymbol{x}} + 0)^d$ we get no lower order effects.

